

Triptesh Acharjee

Roll No.: 234678

B.Sc (Hons) in Physics

Fergusson College (Autonomous), Pune

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📍 Triptesh Acharjee

EDUCATION

•Fergusson College (Autonomous), Pune	2023–2027 (expected)
B.Sc (Hons) in Physics	CGPA: 7.91/10.00
•Jusco School Kadma, Kadma	2021–2023
Pre-University Studies	Percentage: 63.6%
•Carmel Junior College	2011–2021
Secondary Studies	Percentage: 80%

SKILLS

Languages: English, Hindi, Bengali, Marathi, Punjabi

Programming Tools: Python, Java, HTML, Linux, LaTeX, MySQL

MS Office: MS Word, MS Excel, MS PowerPoint

Soft Skills: Public Speaking, Diplomacy, Decision Making, Adaptability, Delegation, Analytical Thinking, Critical Thinking, Research

Coursework: Mechanics, Thermodynamics, Optics, Electrodynamics, Nuclear Physics, Astrophysics, Cosmology, Quantum Mechanics, Particle Physics, Correlation, Data Analysis, Data Visualisation, Petrology, Optical Properties of Minerals

Areas of Interest: Radio Astronomy, Data Analysis, Astrophotography, Telescope Handling

EXPERIENCE

•BRICS Data Analysis Program	May 2026 – Present
Grader and Mentor	Remote
– Assisting as a grader and mentor for participants in the BRICS Data Analysis initiative.	
– Evaluating hands-on capstone assignments on light curve modeling and astronomical data visualization.	
– Providing technical guidance on time-series analysis, Python libraries (astropy , Lightkurve), and data cleaning workflows.	
•Workshop on Data Analytics by IDIA and BRICS	May 2025 – June 2025
Workshop Attendee	Grade: 89/100
– Learned about the foundations of data analytics, data visualization, cleaning, and preprocessing.	
– Completed a capstone project utilizing Lightkurve on TESS satellite data, securing an 89/100 grade.	
•Workshop on Astrophysical Dust Ices: Insights from Recent Telescopes	March 2025
Workshop Attendee	PRL
– Gained insights into astrophysical dust ices and current observational trends in cosmology.	
•Workshop on Radio Astronomy	March 2025
Workshop Attendee	NCRA
– Explored astronomical data handling from various radio observatories.	
– Learned about modern observational techniques in cosmology and radio interferometry.	
•43rd Meeting of the Astronomical Society of India	February 2025
Attendee & Presenter	ASI
– Attended talks on current trends in Indian astronomy and astrophysics by leading researchers.	
– Presented a research poster titled " <i>Accelerating Astronomy: A Unified Data Access</i> ".	
•Cosmology From Home	June 2024
Workshop Attendee	Online Conference
– Attended specialist lectures on theoretical and computational cosmology.	
– Gained hands-on training with cosmological modeling and data analysis tools.	
•Indian Institute of Remote Sensing (IIRS – ISRO)	January 2024
Workshop Attendee	Dehradun
– Completed training on remote sensing principles, satellite image processing, and geospatial analysis.	

•Frontiers in Physics

2024, 2025

Technical Team In-Charge & Attendee

Fergusson College

- Headed technical operations for event management.
- Attended lectures on optical/radio astronomy, attosecond physics, spectroscopy, and semiconductors.

•IUCAA National Science Day

2025

Volunteer & Workshop Attendee

IUCAA, Pune

- Designed and constructed a 1:50 scale model of the Square Kilometre Array Observatory (SKAO).
- Participated in Q&A sessions with IUCAA scientists covering solar physics, SETI, and astrophysics.

•Astro Club Poster Exhibition

2024, 2025

Volunteer

Fergusson College

- Presented interactive models and posters on planetary exploration missions and electrodynamics.

PROJECTS

•CuO-Based Gas Sensor Fabrication using 3D Printed Substrates

Jan 2026 – Apr 2026

Fergusson College (Autonomous), Pune

Status: Completed

- **Guides:** Prof. Dr. Ashish Yengantiwar, Co-guide: Mr. Bosco V. Ruptake
- **Tools & Technologies:** 3D Printing (Proto-pasta & Electrifi PLA), Electrochemical Deposition, XRD, UV-Vis, FTIR, CV.
- Executed a 4-month research project fabricating copper oxide (CuO)-based sensors for detecting reducing gases (specifically ammonia). Optimized conductive electrodes using 3D-printed Proto-pasta PLA to enhance conductivity. Synthesized CuO via electrochemical deposition and led material characterization (XRD, UV-Vis, FTIR, CV) to analyze structural, optical, and electrochemical properties.

•Ionospheric TEC Analysis during the May 2024 Solar Storm

Aug 2025 – Dec 2025

IIT Indore

Status: Completed

- **Guides:** Prof. Dr. Abhirup Dutta (IIT Indore), Co-guide: Mr. Bhuvnesh Brawar
- **Tools & Technologies:** Python, MATLAB, GNSS-TEC Analysis Software, RINEX Data.
- Investigated ionospheric responses to the G5 geomagnetic storm of May 2024 using GNSS-derived Total Electron Content (TEC) data. Processed multi-station RINEX files across the Indian subcontinent to calculate vertical TEC (VTEC), visualizing spatial/temporal ionospheric anomalies caused by solar weather events.

•Detecting Stellar Variability through Periodogram Modelling using TESS Light Curves

May 2025 – Jul 2025

BRICS / Self-Directed

Status: Completed

- **Tools & Technologies:** Python (matplotlib, pandas, astropy, Lightkurve)
- Analyzed photometric data from the TESS satellite to detect variable star periodicities across BRICS observatory targets. Applied Lomb-Scargle periodograms, outlier filtration, and statistical feature extraction (amplitude, mean flux) for classification.

•Accelerating Astronomy: A Unified Data Access Platform

Nov 2025 – Present

Fergusson College (Autonomous), Pune

Status: Paused

- **Tools & Technologies:** Python
- Developing a central platform to aggregate, organize, and search international astrophysics literature with advanced tagging, query mechanisms, and visual analytics.

•Monitoring Heat Wave Conditions Using WBGT Index

May 2024

Fergusson College (Autonomous), Pune

Status: Completed

- **Tools & Technologies:** WBGT Analysis Software, R
- Recorded and evaluated microclimate heat stress metrics (air/globe/wet-bulb temperatures, dew point) across campus grounds to map thermal exposure trends and propose heat mitigation strategies.

ACHIEVEMENTS

- Poster Exhibition by Astro Club (Frontiers in Physics)

Participant

– Secured **1st Position** for research presentation.
- 2026

Fergusson College
- Poster Exhibition by Astro Club

Participant

– Secured **1st Position** in 2024.
– Secured **3rd Position** in 2023.
- 2023, 2024

Fergusson College
- Scienceporium at Little Flower School

Participant

– Secured **3rd Position** in inter-school science exhibition.
- 2022

Jamshedpur

HOBBIES AND INTERESTS

- Writing: Quotes
- Music: Proficient in playing the Guitar
- Sports: Swimming, Waterpolo, Cricket, Badminton, Tennis, Football, Kabaddi
- Reading: Science Fiction, Fantasy, Non-Fiction, Self-Improvement